VOLUME 4D — QUESTION BANK: QUANTITATIVE / MATHEMATICAL APTITUDE (100 Questions)

> **Reality check:** NID needs Class 8–10 maths in design contexts (scale, area, ratio, percentages). This bank is deliberately set in those contexts. Priority: LOW. Aim for speed and accuracy, then stop. Sections: A. Percentages, ratio, proportion (1–20) · B. Averages, statistics & data (21–32) · C. Units, paper & digital measures (33–45) · D. Time, speed, distance, work (46–58) · E. Geometry & mensuration (59–85) · F. Design-context word problems & puzzles (86–100)

Format: Question → **Answer** → working → *Concept* · *Diff* · *Why NID* · *Mistake*.

A. Percentages, Ratio, Proportion (1–20)

QA-1. What is 25% of 480?
(a) 100 (b) 120 (c) 140 (d) 96
Ans: (b). 480 ÷ 4 = 120. *Concept:* percentage · *Diff:* E · *Why NID:* basics · *Mistake:* none.

QA-2. A ₹1,200 table lamp's price rises by 15%. What is the new price?
(a) ₹1,350 (b) ₹1,380 (c) ₹1,400 (d) ₹1,320
Ans: (b). 1200 × 1.15 = 1380. *Concept:* % increase · *Diff:* E · *Why NID:* everyday · *Mistake:* none.

QA-3. A ₹2,000 item is discounted 10%, and the discounted price is then raised by 10%. What is the final price?
(a) ₹2,000 (b) ₹1,980 (c) ₹2,020 (d) ₹1,800
Ans: (b). 2000 × 0.9 × 1.1 = 1980, a net change of −1%. *Concept:* successive % · *Diff:* M · *Why NID:* the classic trap · *Mistake:* assuming the two changes cancel out.

QA-4. Successive discounts of 20% and 10% equal a single discount of:
(a) 30% (b) 28% (c) 25% (d) 32%
Ans: (b). 1 − 0.8 × 0.9 = 0.28. *Concept:* successive % · *Diff:* M · *Why NID:* reasoning · *Mistake:* adding them to get 30%.

QA-5. By what percentage does a price rising from ₹80 to ₹100 increase?
(a) 20% (b) 25% (c) 80% (d) 15%
Ans: (b). 20 ÷ 80 = 25%. *Concept:* % change · *Diff:* E · *Why NID:* data reading · *Mistake:* dividing by the new value (20 ÷ 100 = 20%).

QA-6. By what percentage does a price falling from ₹100 to ₹80 decrease?
(a) 20% (b) 25% (c) 80% (d) 15%
Ans: (a). 20 ÷ 100 = 20%. *Concept:* % change · *Diff:* E · *Why NID:* contrast with QA-5 · *Mistake:* none.

QA-7. 12 is what percentage of 48?
(a) 20% (b) 25% (c) 30% (d) 40%
Ans: (b). *Concept:* % · *Diff:* E · *Why NID:* basics · *Mistake:* none.

QA-8. Red and white paint are mixed in the ratio 2 : 3 to make 10 L of pink. How much red paint is used?
(a) 3 L (b) 4 L (c) 5 L (d) 6 L
Ans: (b). 10 × 2/5 = 4. *Concept:* ratio · *Diff:* E · *Why NID:* colour mixing context · *Mistake:* none.

QA-9. ₹64 is divided in the ratio 3 : 5. What is the larger share?
(a) ₹24 (b) ₹40 (c) ₹38 (d) ₹32
Ans: (b). 64 × 5/8 = 40. *Concept:* ratio · *Diff:* E · *Why NID:* basics · *Mistake:* none.

QA-10. If a : b = 2 : 3 and b : c = 4 : 5, find a : b : c.
(a) 8 : 12 : 15 (b) 2 : 3 : 5 (c) 4 : 6 : 5 (d) 8 : 12 : 10
Ans: (a). Make b equal in both ratios: a : b = 8 : 12 and b : c = 12 : 15. *Concept:* compound ratio · *Diff:* M · *Why NID:* proportion · *Mistake:* answering (b).

QA-11. On a 1 : 200 site plan, a wall measures 4.5 cm. What is its real length?
(a) 4.5 m (b) 9 m (c) 90 m (d) 0.9 m
Ans: (b). 4.5 × 200 = 900 cm = 9 m. *Concept:* scale · *Diff:* E · *Why NID:* architecture drawings · *Mistake:* converting the units wrongly.

QA-12. On a map where 1 cm = 5 km, two towns are 12.4 cm apart. What is the real distance?
(a) 60 km (b) 62 km (c) 64 km (d) 6.2 km
Ans: (b). *Concept:* map scale · *Diff:* E · *Why NID:* scale · *Mistake:* none.

QA-13. A 1 : 10 model of a chair is made. The real seat height is 45 cm. What is the model seat height?
(a) 45 mm (b) 4.5 mm (c) 450 mm (d) 4.5 m
Ans: (a). 45 cm ÷ 10 = 4.5 cm = 45 mm. *Concept:* model scale · *Diff:* E · *Why NID:* Studio Test models · *Mistake:* mixing up cm and mm.

QA-14. A table top measures 50 cm² on a 1 : 20 model. What is the real area?
(a) 1,000 cm² (b) 2 m² (c) 0.2 m² (d) 20 m²
Ans: (b). Areas scale by 20² = 400: 50 × 400 = 20,000 cm² = 2 m². *Concept:* area scaling · *Diff:* M · *Why NID:* scale reasoning · *Mistake:* multiplying by 20 only.

QA-15. If every length of a box is doubled, its volume becomes:
(a) 2× (b) 4× (c) 6× (d) 8×
Ans: (d). $2^3 = 8$. *Concept:* volume scaling · *Diff:* E · *Why NID:* packaging · *Mistake:* answering 2×.

QA-16. Five artisans make a set of baskets in 12 days. How long would 6 artisans take at the same rate?
(a) 10 days (b) 14 days (c) 11 days (d) 9 days
Ans: (a). Total work = 5 × 12 = 60 artisan-days; 60 ÷ 6 = 10. *Concept:* inverse proportion · *Diff:* E · *Why NID:* planning · *Mistake:* using direct proportion.

QA-17. 3 pencils cost ₹12. How much do 15 pencils cost?
(a) ₹45 (b) ₹60 (c) ₹50 (d) ₹36
Ans: (b). *Concept:* direct proportion · *Diff:* E · *Why NID:* basics · *Mistake:* none.

QA-18. A stall buys a mug for ₹400 and sells it for ₹500. What is the profit percentage?
(a) 20% (b) 25% (c) 10% (d) 100%
Ans: (b). Profit % is calculated on the cost price: 100 ÷ 400 = 25%. *Concept:* profit · *Diff:* E · *Why NID:* basics · *Mistake:* calculating on the selling price (20%).

QA-19. A rectangle's length increases by 20% and its width by 10%. By what percentage does its area increase?
(a) 30% (b) 32% (c) 28% (d) 22%
Ans: (b). 1.2 × 1.1 = 1.32. *Concept:* combined % · *Diff:* M · *Why NID:* resizing artwork · *Mistake:* adding to 30%.

QA-20. A poster's width : height is 3 : 4. If the width is 45 cm, what is the height?
(a) 55 cm (b) 60 cm (c) 50 cm (d) 64 cm
Ans: (b). 45 × 4/3 = 60. *Concept:* aspect ratio · *Diff:* E · *Why NID:* layout · *Mistake:* none.

B. Averages, Statistics & Data Interpretation (21–32)

QA-21. Find the average of 12, 15, 18, 21, 24.
(a) 17 (b) 18 (c) 19 (d) 20
Ans: (b). The numbers are evenly spaced, so the average is the middle value, 18. *Concept:* average · *Diff:* E · *Why NID:* basics · *Mistake:* none.

QA-22. Six numbers average 20. When one number is removed, the remaining five average 18. What number was removed?
(a) 28 (b) 30 (c) 32 (d) 20
Ans: (b). 6 × 20 − 5 × 18 = 120 − 90 = 30. *Concept:* average manipulation · *Diff:* M · *Why NID:* reasoning · *Mistake:* none.

QA-23. A student scored 70 and 80 in two tests. What score in the third test gives an average of 80?
(a) 85 (b) 90 (c) 95 (d) 80
Ans: (b). 240 − 150 = 90. *Concept:* target average · *Diff:* E · *Why NID:* planning · *Mistake:* none.

QA-24. For the data 2, 3, 3, 5, 7, find the mean, median and mode.
(a) 4, 3, 3 (b) 4, 5, 3 (c) 3, 3, 4 (d) 5, 3, 3
Ans: (a). Mean = 20 ÷ 5 = 4; median = the middle value = 3; mode = the most frequent = 3. *Concept:* central tendency · *Diff:* E · *Why NID:* survey analysis · *Mistake:* none.

QA-25. Find the range of 4, 9, 2, 15.
(a) 11 (b) 13 (c) 15 (d) 6
Ans: (b). 15 − 2 = 13. *Concept:* range · *Diff:* E · *Why NID:* data · *Mistake:* none.

QA-26. In a pie chart, a category is 25% of the total. What angle does its slice make?
(a) 25° (b) 90° (c) 45° (d) 100°
Ans: (b). 0.25 × 360 = 90°. *Concept:* pie chart · *Diff:* E · *Why NID:* infographics · *Mistake:* none.

QA-27. A pie-chart slice measures 72°. What percentage does it represent?
(a) 18% (b) 20% (c) 25% (d) 72%
Ans: (b). 72 ÷ 360 = 0.2. *Concept:* pie chart · *Diff:* E · *Why NID:* infographics · *Mistake:* none.

QA-28. In a survey of 50 students, 60% like drawing. How many like drawing?
(a) 25 (b) 30 (c) 35 (d) 20
Ans: (b). *Concept:* % of a total · *Diff:* E · *Why NID:* research · *Mistake:* none.

QA-29. A bar chart shows the visitors to a craft fair: Mon 200, Tue 300, Wed 250, Thu 450, Fri 300. On which day did visitors increase the most compared with the previous day?
(a) Tue (b) Wed (c) Thu (d) Fri
Ans: (c). The day-on-day changes are +100, −50, +200, −150. *Concept:* data reading · *Diff:* E · *Why NID:* data visualisation · *Mistake:* choosing the highest absolute value instead of the biggest change.

QA-30. In QA-29, what is the average number of visitors per day?
(a) 280 (b) 300 (c) 250 (d) 320
Ans: (b). Total 1500 ÷ 5 = 300. *Concept:* average from a chart · *Diff:* E · *Why NID:* data · *Mistake:* none.

QA-31. A designer estimated 95 units but the actual number was 100. What is the percentage error, relative to the actual?
(a) 5% (b) 5.26% (c) 95% (d) 10%
Ans: (a). 5 ÷ 100 = 5%. *Concept:* error % · *Diff:* E · *Why NID:* estimation · *Mistake:* dividing by the estimate.

QA-32. A class's marks are: 10 students scored 60 and 30 students scored 80. What is the class average?
(a) 70 (b) 75 (c) 72 (d) 78
Ans: (b). (600 + 2400) ÷ 40 = 75. *Concept:* weighted average · *Diff:* M · *Why NID:* reasoning · *Mistake:* taking the simple average of 60 and 80 (70).

C. Units, Paper & Digital Measures (33–45)

QA-33. How many centimetres are in 2.5 m?
(a) 25 (b) 250 (c) 2,500 (d) 0.25
Ans: (b). *Concept:* units · *Diff:* E · *Why NID:* measurement · *Mistake:* none.

QA-34. 10 inches is approximately:
(a) 2.54 cm (b) 25.4 cm (c) 254 cm (d) 30.5 cm
Ans: (b). *Concept:* imperial–metric · *Diff:* E · *Why NID:* product sizes (TV screens, for example) · *Mistake:* none.

QA-35. A person is 6 feet tall. Approximately how tall is that in cm?
(a) 172 (b) 183 (c) 190 (d) 160
Ans: (b). 6 × 30.48 = 182.88. *Concept:* units · *Diff:* E · *Why NID:* anthropometry · *Mistake:* none.

QA-36. What is the area of an A0 sheet?
(a) 0.5 m² (b) 1 m² (c) 2 m² (d) 0.25 m²
Ans: (b). A0 is defined as 1 m² with sides in the ratio 1 : √2 (841 × 1189 mm). *Concept:* ISO paper · *Diff:* M · *Why NID:* design materials · *Mistake:* none.

QA-37. How many A4 sheets have the same area as one A1 sheet?
(a) 4 (b) 6 (c) 8 (d) 16
Ans: (c). A1 = 2 A2 = 4 A3 = 8 A4. *Concept:* paper sizes · *Diff:* E · *Why NID:* design practice · *Mistake:* answering 4.

QA-38. The long side ÷ short side ratio of any A-series sheet is about:
(a) 1.5 (b) 1.414 (c) 1.618 (d) 1.333
Ans: (b). √2. This ratio means that halving a sheet keeps the same proportions. *Concept:* the √2 ratio · *Diff:* M · *Why NID:* proportion · *Mistake:* answering 1.618 (the golden ratio).

QA-39. How many pixels does a screen of 1920 × 1080 have?
(a) about 2.07 million (b) about 1 million (c) about 20.7 million (d) 3,000
Ans: (a). 1920 × 1080 = 2,073,600. *Concept:* digital measure · *Diff:* E · *Why NID:* digital design · *Mistake:* none.

QA-40. What is the aspect ratio of 1920 × 1080?
(a) 4 : 3 (b) 16 : 9 (c) 3 : 2 (d) 21 : 9
Ans: (b). Divide both by 120. *Concept:* aspect ratio · *Diff:* E · *Why NID:* screen and video design · *Mistake:* none.

QA-41. A 4 × 6 inch photo is printed at 300 dpi. What pixel size is needed?
(a) 400 × 600 (b) 1200 × 1800 (c) 300 × 450 (d) 2400 × 3600
Ans: (b). Multiply each side by 300. *Concept:* print resolution · *Diff:* M · *Why NID:* graphic production · *Mistake:* none.

QA-42. In typography 72 points ≈ 1 inch. What is the height of 36 pt type?
(a) 0.25 in (b) 0.5 in (c) 1 in (d) 2 in
Ans: (b). *Concept:* type measure · *Diff:* E · *Why NID:* typography · *Mistake:* none.

QA-43. How many litres does a tank measuring 1 m × 1 m × 1 m hold?
(a) 100 (b) 1,000 (c) 10,000 (d) 10
Ans: (b). *Concept:* volume–capacity · *Diff:* E · *Why NID:* product sizing · *Mistake:* none.

QA-44. How many square metres are in 1 hectare?
(a) 1,000 (b) 10,000 (c) 100,000 (d) 100
Ans: (b). *Concept:* area units · *Diff:* E · *Why NID:* landscape/campus scale · *Mistake:* none.

QA-45. India Standard Time is UTC + 5:30. What is the IST time at 12:00 noon UTC?
(a) 6:30 pm (b) 5:30 pm (c) 6:30 am (d) 5:30 am
Ans: (b). *Concept:* time zones · *Diff:* E · *Why NID:* general awareness · *Mistake:* none.

D. Time, Speed, Distance, Work (46–58)

QA-46. How long does it take to cover 150 km at 60 km/h?
(a) 2 h (b) 2.5 h (c) 3 h (d) 1.5 h
Ans: (b). *Concept:* d = st · *Diff:* E · *Why NID:* basics · *Mistake:* none.

QA-47. What is 36 km/h in m/s?
(a) 10 (b) 36 (c) 3.6 (d) 100
Ans: (a). 36 × 5/18 = 10. *Concept:* unit conversion · *Diff:* E · *Why NID:* basics · *Mistake:* none.

QA-48. A 200 m train moving at 72 km/h passes a pole. How long does it take?
(a) 10 s (b) 20 s (c) 2.8 s (d) 15 s
Ans: (a). 72 km/h = 20 m/s, and 200 ÷ 20 = 10 s. *Concept:* train problems · *Diff:* M · *Why NID:* reasoning · *Mistake:* forgetting to convert the units.

QA-49. The same train crosses a 300 m platform. How long does it take?
(a) 15 s (b) 25 s (c) 10 s (d) 30 s
Ans: (b). It must cover its own length plus the platform: 500 ÷ 20 = 25 s. *Concept:* train + platform · *Diff:* M · *Why NID:* reasoning · *Mistake:* using only 300 m.

QA-50. A cyclist goes 60 km at 30 km/h and returns at 60 km/h. What is the average speed for the whole trip?
(a) 45 km/h (b) 40 km/h (c) 50 km/h (d) 42 km/h
Ans: (b). Total time = 2 h + 1 h = 3 h for 120 km, so 40 km/h. *Concept:* average speed (a harmonic mean) · *Diff:* M · *Why NID:* the classic trap · *Mistake:* averaging the two speeds (45).

QA-51. Walking at 5 km/h, how long does it take to cover 2.5 km?
(a) 20 min (b) 30 min (c) 45 min (d) 50 min
Ans: (b). *Concept:* time · *Diff:* E · *Why NID:* basics · *Mistake:* none.

QA-52. A can paint a mural in 6 days and B can paint it in 12 days. How long do they take working together?
(a) 9 days (b) 4 days (c) 3 days (d) 18 days
Ans: (b). 1/6 + 1/12 = 3/12 = 1/4 of the job per day. *Concept:* work rates · *Diff:* M · *Why NID:* collaboration problems · *Mistake:* averaging the times.

QA-53. A tap fills a tank in 4 h and a drain empties it in 6 h. With both open, how long does the tank take to fill?
(a) 10 h (b) 12 h (c) 5 h (d) 2.4 h
Ans: (b). 1/4 − 1/6 = 1/12. *Concept:* rates · *Diff:* M · *Why NID:* reasoning · *Mistake:* none.

QA-54. What is the angle between the clock hands at 4:00?
(a) 90° (b) 120° (c) 150° (d) 100°
Ans: (b). 4 × 30° = 120°. *Concept:* clock angles · *Diff:* E · *Why NID:* reasoning · *Mistake:* none.

QA-55. What is the smaller angle between the clock hands at 9:15?
(a) 172.5° (b) 180° (c) 187.5° (d) 165°
Ans: (a). $|30 \times 9 - 5.5 \times 15| = |270 - 82.5| = 187.5°$, so the smaller angle is 360° − 187.5° = 172.5°. *Concept:* clock angles · *Diff:* M · *Why NID:* precision · *Mistake:* reporting the larger angle (187.5°).

QA-56. What is the angle between the clock hands at 3:30?
(a) 90° (b) 75° (c) 60° (d) 105°
Ans: (b). $|90 - 165| = 75°$. *Concept:* clock angles · *Diff:* M · *Why NID:* reasoning · *Mistake:* answering 90°.

QA-57. A bus leaves every 15 minutes from 6:00 am. How many buses leave between 6:00 am and 8:00 am, inclusive of both times?
(a) 8 (b) 9 (c) 10 (d) 7
Ans: (b). 120 ÷ 15 = 8 intervals, so there are 9 departures. *Concept:* fence-post counting · *Diff:* M · *Why NID:* reasoning · *Mistake:* answering 8.

QA-58. A father is 3 times as old as his son. In 10 years he will be twice as old. How old is the son now?
(a) 8 (b) 10 (c) 12 (d) 15
Ans: (b). 3x + 10 = 2(x + 10), so x = 10. *Concept:* linear equations · *Diff:* M · *Why NID:* reasoning · *Mistake:* none.

E. Geometry & Mensuration (59–85)

(Use π = 22/7 where convenient.)

QA-59. Find the area and perimeter of a 12 × 8 rectangle.
(a) 96, 40 (b) 96, 20 (c) 80, 40 (d) 48, 40
Ans: (a). *Concept:* rectangle · *Diff:* E · *Why NID:* basics · *Mistake:* none.

QA-60. What is the area of a circle of radius 7 cm?
(a) 44 cm² (b) 154 cm² (c) 144 cm² (d) 49 cm²
Ans: (b). 22/7 × 49 = 154. *Concept:* circle area · *Diff:* E · *Why NID:* basics · *Mistake:* computing the circumference (44).

QA-61. What is the circumference of a circular table of radius 14 cm?
(a) 44 (b) 88 (c) 616 (d) 28
Ans: (b). 2 × 22/7 × 14 = 88. *Concept:* circumference · *Diff:* E · *Why NID:* edge-banding a table, for example · *Mistake:* none.

QA-62. Find the volume and surface area of a cube of side 5.
(a) 125, 150 (b) 125, 100 (c) 25, 150 (d) 150, 125
Ans: (a). *Concept:* cube · *Diff:* E · *Why NID:* basics · *Mistake:* none.

QA-63. Find the volume and total surface area of a 10 × 5 × 4 cuboid.
(a) 200, 220 (b) 200, 110 (c) 190, 220 (d) 220, 200
Ans: (a). Surface area = 2(50 + 20 + 40) = 220. *Concept:* cuboid · *Diff:* E · *Why NID:* packaging · *Mistake:*

none.

QA-64. A cylindrical bottle has radius 7 cm and height 10 cm. What is its capacity?
(a) 1.54 L (b) 15.4 L (c) 0.154 L (d) 154 L
Ans: (a). 22/7 × 49 × 10 = 1540 cm³ = 1.54 L. *Concept:* cylinder · *Diff:* M · *Why NID:* product design · *Mistake:* converting the units wrongly.

QA-65. A cone has radius 3 and height 4. What is its slant height?
(a) 5 (b) 7 (c) 25 (d) 12
Ans: (a). √(9 + 16) = 5. *Concept:* Pythagoras in 3D · *Diff:* E · *Why NID:* packaging (cones) · *Mistake:* none.

QA-66. What is the curved surface area of that cone (r = 3, l = 5)?
(a) 15π (b) 12π (c) 9π (d) 30π
Ans: (a). πrl = 15π ≈ 47.1. *Concept:* cone surface · *Diff:* M · *Why NID:* paper cone cups · *Mistake:* using the height instead of the slant height.

QA-67. What is the volume of a sphere of radius 3?
(a) 36π (b) 27π (c) 12π (d) 108π
Ans: (a). (4/3)π × 27 = 36π. *Concept:* sphere · *Diff:* M · *Why NID:* form · *Mistake:* none.

QA-68. What is the area of a triangle with base 10 and height 6?
(a) 60 (b) 30 (c) 16 (d) 36
Ans: (b). *Concept:* triangle · *Diff:* E · *Why NID:* basics · *Mistake:* forgetting the ½.

QA-69. A 5 m × 4 m room is tiled with 25 cm × 25 cm tiles. How many tiles are needed?
(a) 200 (b) 320 (c) 400 (d) 80
Ans: (b). 500 × 400 ÷ 625 = 320. *Concept:* area division · *Diff:* E · *Why NID:* interior design · *Mistake:* mixing m and cm.

QA-70. A 3 m × 4 m wall needs two coats of paint, and 1 L covers 6 m² per coat. How much paint is needed?
(a) 2 L (b) 4 L (c) 6 L (d) 3 L
Ans: (b). 12 m² × 2 coats = 24 m², and 24 ÷ 6 = 4 L. *Concept:* applied area · *Diff:* E · *Why NID:* practical estimation · *Mistake:* forgetting the second coat.

QA-71. What is the interior angle of a regular pentagon?
(a) 100° (b) 108° (c) 120° (d) 72°
Ans: (b). (5 − 2) × 180 ÷ 5 = 108. *Concept:* polygon angles · *Diff:* E · *Why NID:* pattern geometry · *Mistake:* giving the exterior angle (72°).

QA-72. What is the sum of the interior angles of a hexagon?
(a) 540° (b) 720° (c) 900° (d) 360°
Ans: (b). *Concept:* angle sum · *Diff:* E · *Why NID:* geometry · *Mistake:* none.

QA-73. What is each exterior angle of a regular octagon?
(a) 30° (b) 45° (c) 135° (d) 60°
Ans: (b). 360 ÷ 8. *Concept:* exterior angles · *Diff:* E · *Why NID:* geometry · *Mistake:* giving the interior angle (135°).

QA-74. What is the diagonal of a square of side 10?
(a) 10 (b) 14.14 (c) 20 (d) 12
Ans: (b). 10√2 ≈ 14.14. *Concept:* Pythagoras · *Diff:* E · *Why NID:* geometry · *Mistake:* none.

QA-75. What is the diagonal of a 6 × 8 rectangle?
(a) 10 (b) 14 (c) 12 (d) 9
Ans: (a). *Concept:* Pythagoras · *Diff:* E · *Why NID:* geometry · *Mistake:* none.

QA-76. Circles have radii in the ratio 2 : 3. What is the ratio of their areas?
(a) 2 : 3 (b) 4 : 9 (c) 8 : 27 (d) 4 : 6
Ans: (b). *Concept:* area scaling · *Diff:* E · *Why NID:* scale · *Mistake:* none.

QA-77. What is the area of a trapezium with parallel sides 8 and 12 and height 5?
(a) 40 (b) 50 (c) 60 (d) 100
Ans: (b). ½ × 20 × 5 = 50. *Concept:* trapezium · *Diff:* E · *Why NID:* geometry · *Mistake:* none.

QA-78. A ring (annulus) has outer radius 7 and inner radius 3.5. What is its area?
(a) 115.5 (b) 154 (c) 38.5 (d) 77
Ans: (a). π(49 − 12.25) = 22/7 × 36.75 = 115.5. *Concept:* annulus · *Diff:* M · *Why NID:* product forms (washers, bangles) · *Mistake:* subtracting the radii before squaring.

QA-79. What is the perimeter of a semicircle of radius 7?
(a) 22 (b) 36 (c) 44 (d) 29
Ans: (b). Curved part + diameter: πr + 2r = 22 + 14 = 36. *Concept:* composite perimeter · *Diff:* M · *Why NID:* geometry · *Mistake:* forgetting the diameter.

QA-80. If a sphere's radius doubles, its volume becomes:
(a) 2× (b) 4× (c) 8× (d) 6×
Ans: (c). *Concept:* scaling · *Diff:* E · *Why NID:* scale · *Mistake:* answering 4×.

QA-81. What is the area of an equilateral triangle of side 6?
(a) 9√3 ≈ 15.6 (b) 18 (c) 36 (d) 12√3
Ans: (a). (√3/4) × 36 = 9√3. *Concept:* special triangles · *Diff:* M · *Why NID:* geometry · *Mistake:* none.

QA-82. What is the area of a regular hexagon of side 2?
(a) 6√3 ≈ 10.4 (b) 12 (c) 4√3 (d) 24
Ans: (a). It is made of 6 equilateral triangles: 6 × (√3/4) × 4 = 6√3. *Concept:* hexagon area · *Diff:* M · *Why NID:* tiling · *Mistake:* none.

QA-83. An open box is made from a 20 cm × 20 cm sheet by cutting a 5 cm square from each corner and folding up the sides. What is its volume?
(a) 500 cm³ (b) 1,000 cm³ (c) 2,000 cm³ (d) 400 cm³
Ans: (a). The box is 10 × 10 × 5. *Concept:* nets to volume · *Diff:* M · *Why NID:* packaging · *Mistake:* using 20 − 5 instead of 20 − 2 × 5 for the base.

QA-84. What is the largest square tile that can exactly cover a 48 cm × 36 cm board, and how many such tiles are needed?
(a) 12 cm, 12 tiles (b) 6 cm, 48 tiles (c) 12 cm, 16 tiles (d) 4 cm, 108 tiles
Ans: (a). HCF(48, 36) = 12, giving a 4 × 3 grid of 12 tiles. *Concept:* HCF · *Diff:* M · *Why NID:* tiling layouts · *Mistake:* none.

QA-85. One border motif repeats every 6 cm and another every 8 cm. After how many cm do they line up again?
(a) 14 (b) 24 (c) 48 (d) 12
Ans: (b). LCM(6, 8) = 24. *Concept:* LCM · *Diff:* E · *Why NID:* textile pattern repeats · *Mistake:* none.

F. Design-Context Word Problems & Puzzles (86–100)

QA-86. Twelve 50 × 50 cm squares are needed for six cushion covers (front and back). Fabric comes 2 m wide. What length of fabric is needed, ignoring seams and waste?
(a) 1 m (b) 1.5 m (c) 2 m (d) 3 m
Ans: (b). 4 squares fit across the 2 m width, so 12 squares need 3 rows × 0.5 m = 1.5 m. *Concept:* layout optimisation · *Diff:* M · *Why NID:* textile planning · *Mistake:* dividing the area (3 m² ÷ 2 m) without checking that the squares actually fit, which happens to give the same answer here.

QA-87. A rectangle has width 10. What length makes it a golden rectangle?
(a) 14.1 (b) 16.18 (c) 15 (d) 20
Ans: (b). 10 × 1.618. *Concept:* golden ratio · *Diff:* E · *Why NID:* proportion · *Mistake:* using √2.

QA-88. Fencing a 20 m straight side with posts every 2 m, including both ends, needs how many posts?
(a) 10 (b) 11 (c) 12 (d) 9
Ans: (b). *Concept:* fence-post problem · *Diff:* E · *Why NID:* reasoning · *Mistake:* answering 10.

QA-89. How many cuts are needed to cut a 10 m rope into 1 m pieces?
(a) 10 (b) 9 (c) 11 (d) 5
Ans: (b). *Concept:* cuts = pieces − 1 · *Diff:* E · *Why NID:* reasoning · *Mistake:* answering 10.

QA-90. A staircase rises 3 m with risers of 15 cm. How many risers are needed?
(a) 15 (b) 20 (c) 25 (d) 30
Ans: (b). 300 ÷ 15. *Concept:* architectural maths · *Diff:* E · *Why NID:* ergonomics · *Mistake:* none.

QA-91. A wheelchair ramp must have a slope of 1 : 12. How long must it run horizontally to rise 60 cm?
(a) 5 m (b) 7.2 m (c) 6 m (d) 12 m
Ans: (b). 60 × 12 = 720 cm. *Concept:* slope ratio · *Diff:* E · *Why NID:* accessible design · *Mistake:* none.

QA-92. Find 3/4 of 2/3 of 120.
(a) 60 (b) 80 (c) 90 (d) 40
Ans: (a). 120 × 2/3 = 80, and 80 × 3/4 = 60. *Concept:* fractions · *Diff:* E · *Why NID:* basics · *Mistake:* none.

QA-93. Write 0.125 as a fraction.
(a) 1/4 (b) 1/8 (c) 1/6 (d) 1/5
Ans: (b). *Concept:* decimals · *Diff:* E · *Why NID:* basics · *Mistake:* none.

QA-94. A die is rolled. What is the probability of an even number?
(a) 1/3 (b) 1/2 (c) 1/6 (d) 2/3
Ans: (b). *Concept:* probability · *Diff:* E · *Why NID:* basics · *Mistake:* none.

QA-95. A bag holds 3 red and 5 blue beads. What is the probability of drawing red?
(a) 3/5 (b) 3/8 (c) 5/8 (d) 1/3
Ans: (b). *Concept:* probability · *Diff:* E · *Why NID:* basics · *Mistake:* answering 3/5 (red : blue).

QA-96. What is the sum of the first 10 natural numbers?
(a) 45 (b) 55 (c) 50 (d) 100
Ans: (b). n(n + 1)/2 = 55. *Concept:* series sum · *Diff:* E · *Why NID:* pattern · *Mistake:* none.

QA-97. What is the sum of the first 10 odd numbers?
(a) 50 (b) 100 (c) 99 (d) 121
Ans: (b). The sum of the first n odd numbers is n² (picture growing square layers). *Concept:* visual number pattern · *Diff:* M · *Why NID:* links to visual series · *Mistake:* none.

QA-98. A Prelims paper has 3 hours. If 25 minutes go to 22 MCQs, 10 minutes to reading and 10 minutes to a buffer, how many minutes per mark remain for a 70-mark subjective section?
(a) about 1.5 (b) about 1.93 (c) about 2.5 (d) about 3
Ans: (b). 180 − 45 = 135 min, and 135 ÷ 70 ≈ 1.93 min per mark, so a 20-mark question gets about 38–39 min. *Concept:* time allocation · *Diff:* M · *Why NID:* exam strategy · *Mistake:* none.

QA-99. 37 °C equals how many °F?
(a) 96.8 (b) 98.6 (c) 100.4 (d) 99.5
Ans: (b). 37 × 9/5 + 32 = 98.6. *Concept:* temperature conversion · *Diff:* E · *Why NID:* general awareness · *Mistake:* none.

QA-100. What is the ratio of consecutive Fibonacci numbers 21 and 13 (21/13), to two decimals, and what does it approach?
(a) 1.62, the golden ratio (b) 1.41, √2 (c) 1.50, 3/2 (d) 1.73, √3
Ans: (a). 21 ÷ 13 ≈ 1.615, and the ratio approaches φ ≈ 1.618. *Concept:* Fibonacci and the golden ratio · *Diff:* M · *Why NID:* design and maths link · *Mistake:* none.

NEXT: Volume 4E — Verbal Ability (100 questions) (`04E-VOL4-QB-VERBAL.md`)